Introduction to Machine Learning -1-

Requirements:

- Practice small tests : 30%
- Practice solutions (pairs: 70%)
- End of semester test: 40%
- Presentation of a project (pairs)

Grade: Exam 60 points + project 100 points

85% - 70% - 55% - 40%

Subjects:
- kNN
- Decision tree
- Ensembles / random forest
- Lin / log regression
- SVM
- FFNN
- CNN
- PCA, autoencoder
- Maximum likelihood
- k means, Gaussian mixture
- Genetic algorithm

Learning:

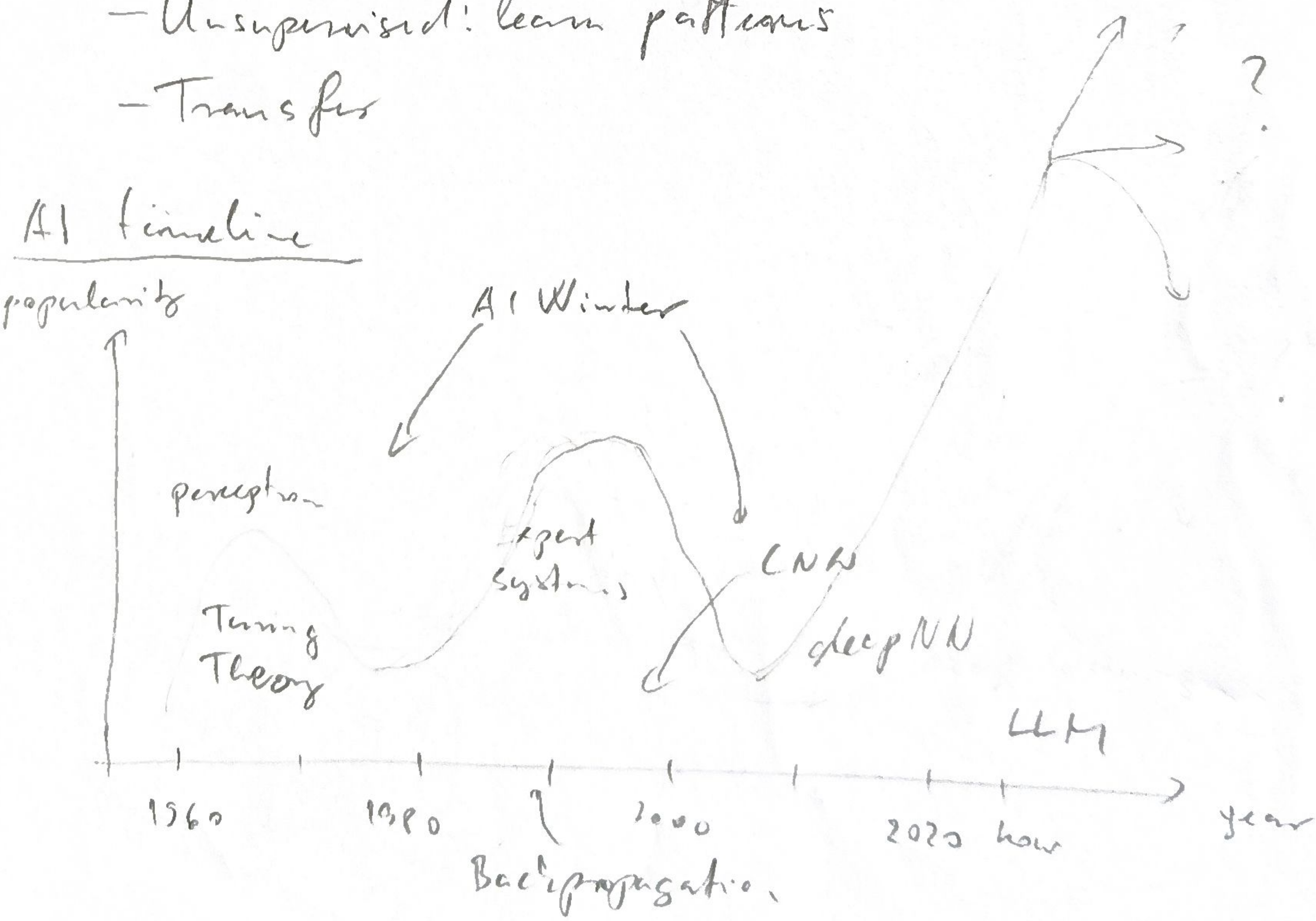
- process of gaining knowledge
- learn from experience E, with respect to task T and performance measure P
- learn a lot ≠ being intelligent

machine learning ≠ artificial intelligence

Learning types:

- Supervised: labelled training data
- Reinforcement: maximize reward based on performance
- Unsupervised: learn patterns
- Transfer

AI timeline



IML -1-

Supervised learning:

training set: (input, labels) $T = (\underline{X}, t)$

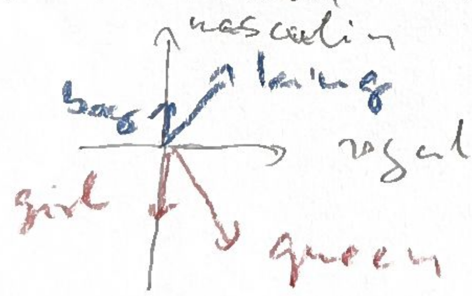
Goal predict labels for data not in training set

Input vector: $\underline{X} \in \mathbb{R}^d$

Representation:

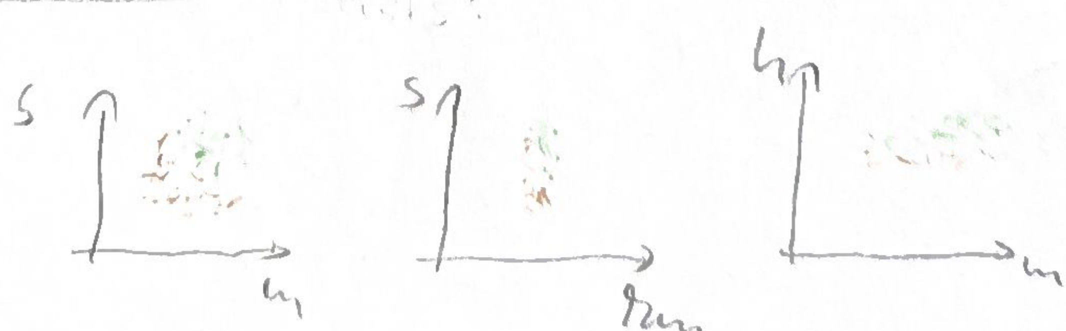
- numerical data: as is
- image: pixels as RGB
- sound: spectrogram, Fourier spectrum as image
- objects: one hot encoding:

  e.g. cat 100 For d different objects
  dog 010 d dimensional vector
  rat 001
- words: embedded vector (incorporating it's meaning)



Numerical data:

Units:



Usual normalization

$$\tilde{X}_\alpha^{(i)} = \frac{x_\alpha^{(i)} - \mu_\alpha}{\sigma_\alpha} \rightarrow \text{zero mean, unit variance}$$

Distance

Problem: points in large dimension are far from each other

e.g. random points on unit sphere

$u, v \in S^{d-1}$

$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2\langle u,v\rangle = 2 - 2\langle u,v\rangle$$

$$z = \langle u,v\rangle = \sum_i u_i v_i \qquad \mathbb{E}[z] = 0 \quad \mathrm{Var}(z) = \frac{1}{d}$$

$$\mathbb{E}[\|u-v\|] = \sqrt{2} - \mathcal{O}\left(\frac{1}{\sqrt{d}}\right)$$

Solution: different distance measures

- Euclidean: $\| \underline{x}^{(i)}, \underline{x}^{(j)} \| = \sqrt{\sum_{\alpha=1}^{d} \left(x_\alpha^{(i)} - x_\alpha^{(j)}\right)^2}$
- Hamming: $H(\underline{x}^{(i)}, \underline{x}^{(j)}) = d - \sum_{i=1}^{d} \delta(x_\alpha^{(i)}, x_\alpha^{(j)})$
- Cosine: $1 - \frac{\langle x^{(i)}, x^{(j)}\rangle}{\|x^{(i)}\| \cdot \|x^{(j)}\|}$

Target or label types

- Regression: $t \in \mathbb{R}$
- Classification: $t \in \{1, \ldots, C\}$ discrete

k - nearest neighbour

(k=1 case)

For a test point x^* find the closest training point. Return its label

$N = |T|$ the number of training points

$$i = \arg\min_{j=1 \dots N} dist(x^*, x^{(j)})$$

$$y = t^{(i)}$$

Voronoi - tesselation :

$$R_k = \{ \underline{x} \in X \mid d(\underline{x}, P_k) \leq d(\underline{x}, P_j) \; \forall j \neq k \}$$

↑ domain of point P_k

Decision boundary

problem:

k-nearest neighbor: majority of the k nearest training points

Small k

- fine pattern
- overfitting

Large k

- stable pattern
- underfitting

popular choice : $k < \sqrt{N}$

Hyperparameter

(here k)

- a parameter the algorithm cannot fit
- controls the behavior of the algorithm

Optimization of the hyperparameter

$\boxed{\text{DATA}}$

$\boxed{\text{train}}$ ~70%

$\boxed{\text{validation}}$ ~20%

$\boxed{\text{test}}$ ~10%

Summary of kNN

- training time : $\emptyset$
- test time :
  - distance : $d \cdot N$
  - sort : $N \log(N)$

→ best for low dim. and small training set